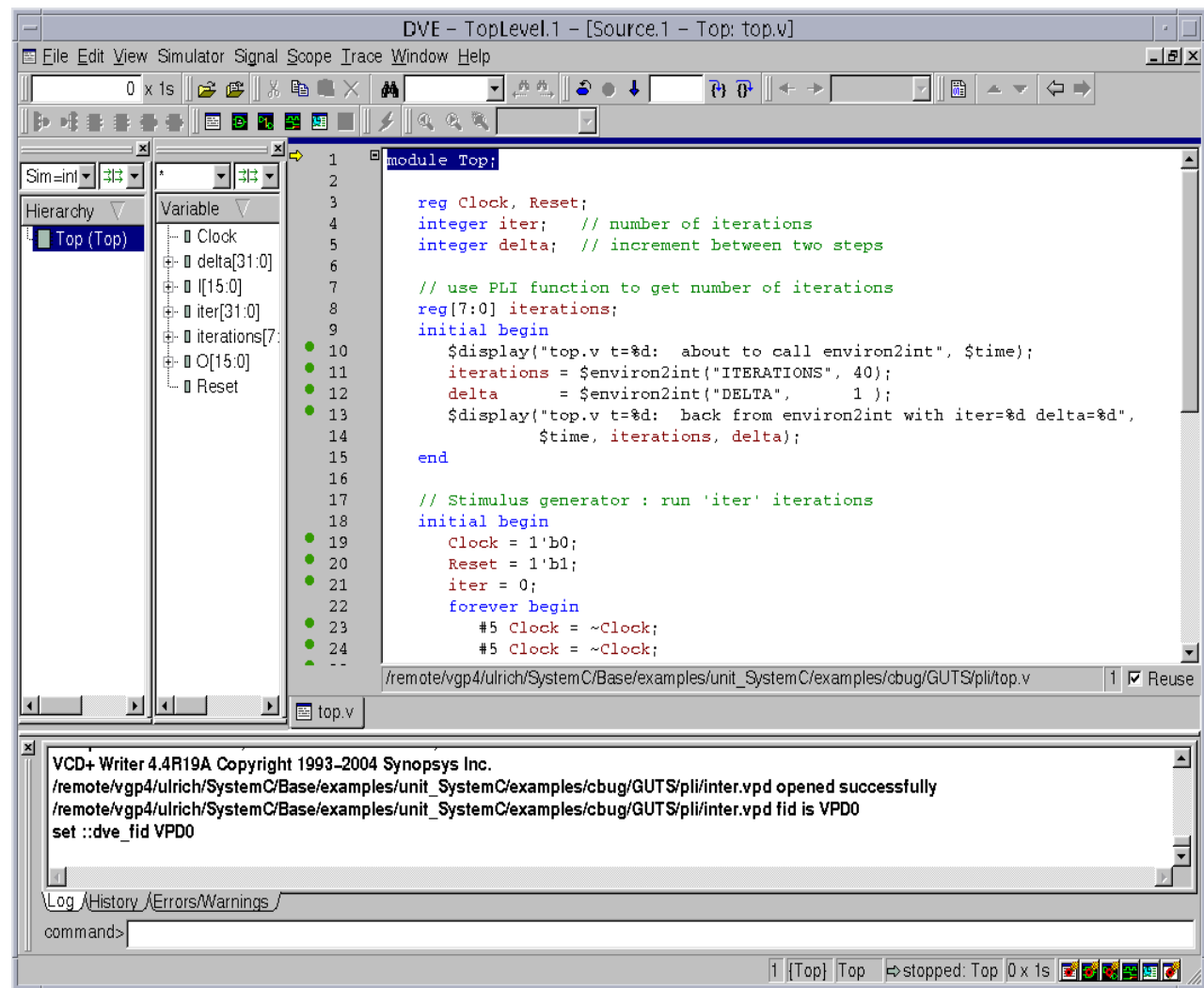


PLI Debugging with DVE / CBug

Compile example
with
comp.csh

Start DVE
simv -gui

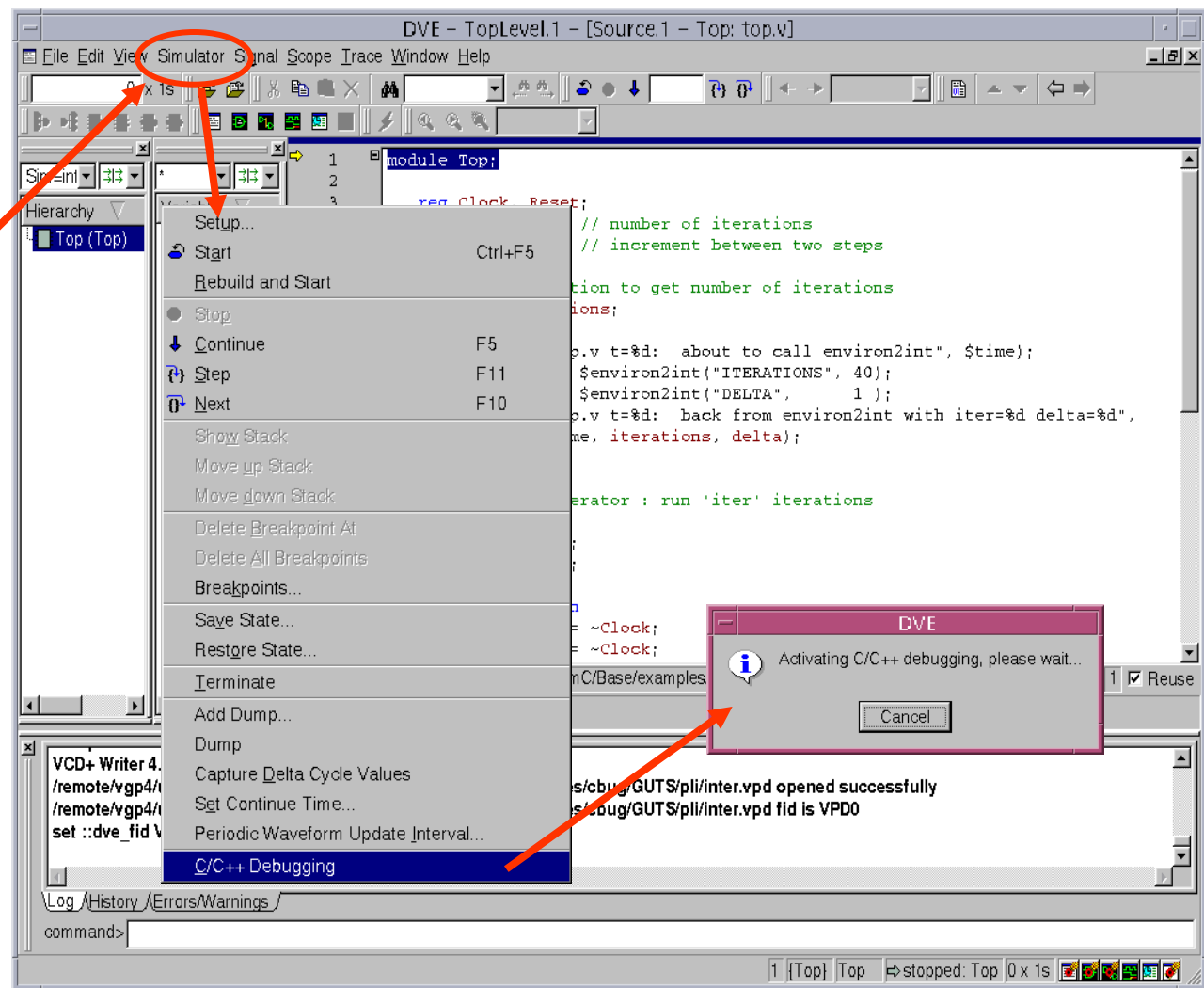


Enable CBug with
menu

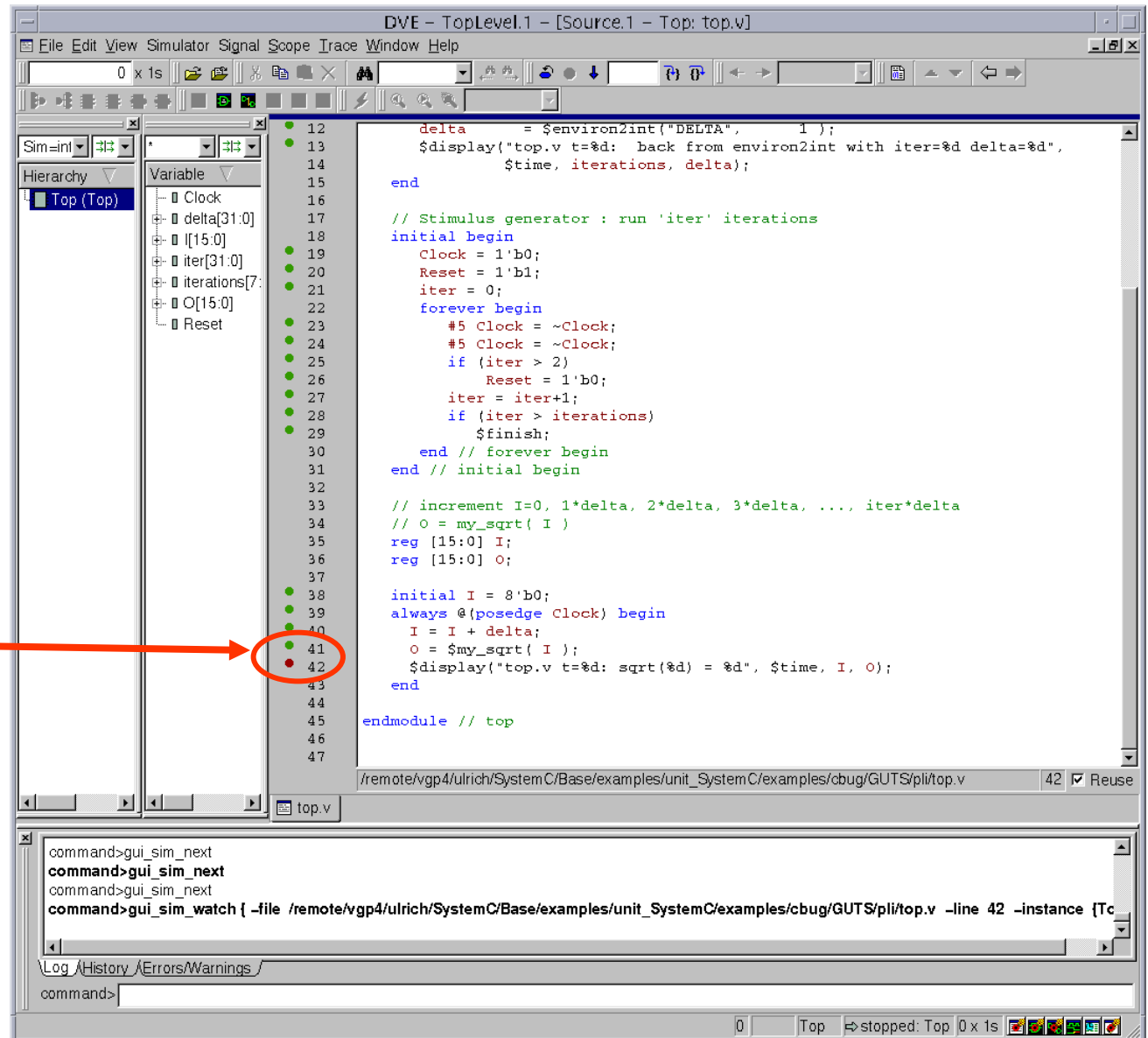
Simulator / C/C++ Debugging

This will take a
few seconds.

Note: menu item
has a check mark
afterwards.



Set
breakpoint in
line 42 of
top.v



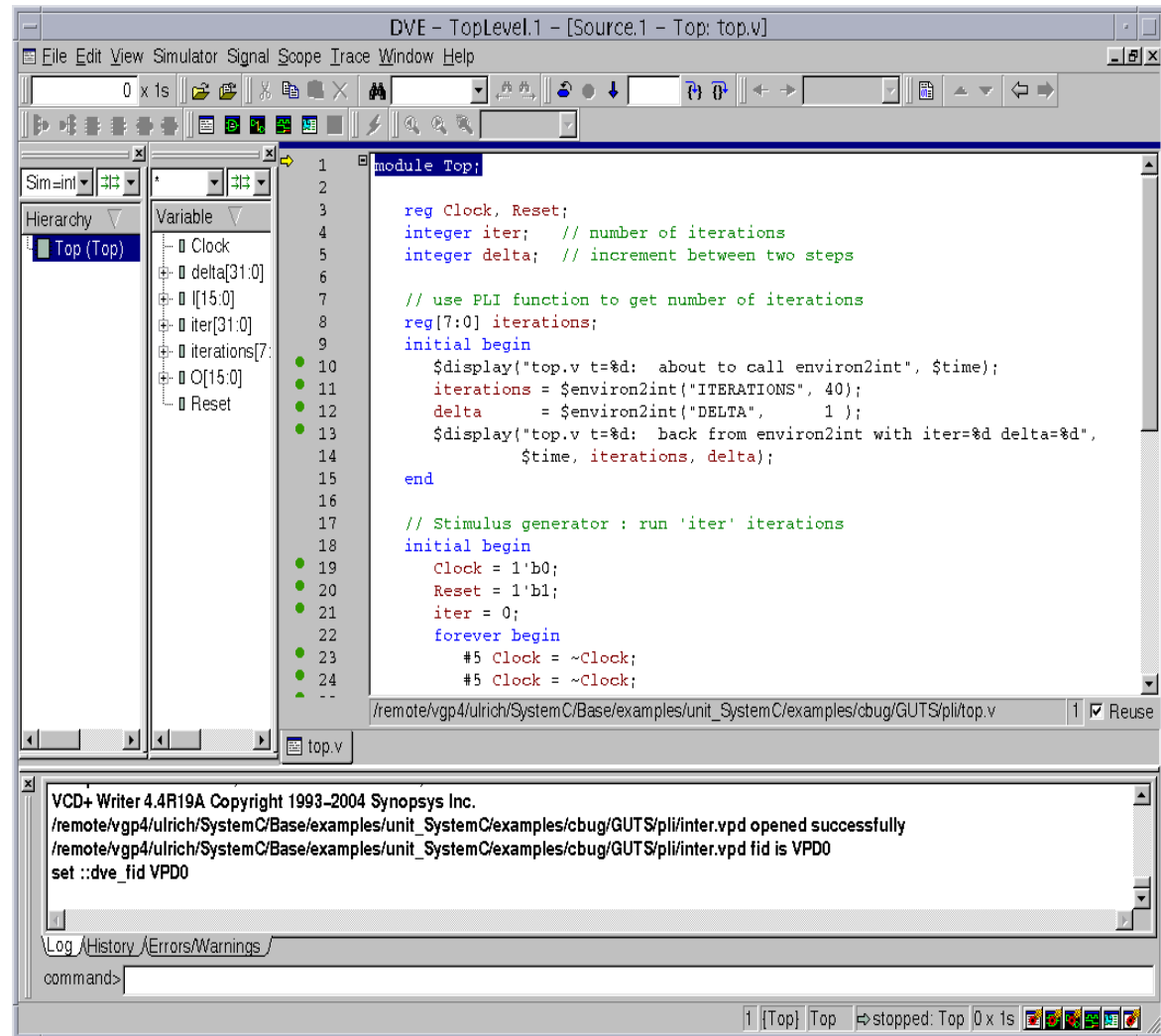
There multiple ways to make DVE stop in C source code:

(A) *cross-step* from Verilog into C

→ Currently only supported for the “call” function of PLI

(B) *Open C source file* and set breakpoint

→ Always works: PLI, DirectC, VhPI, MHPI, ...



(A) cross-step from Verilog into C

Step (F11)

Verilog

```
1 module Top;
2
3 reg Clock, Reset;
4 integer iter; // number of iterations
5 integer delta; // increment between two steps
6
7 // use PLI function to get number of iterations
8 reg[7:0] iterations;
9 initial begin
10 $display("top.v t=%d: about to call environ2int", $time);
11 iterations = $environ2int("ITERATIONS", 40);
12 delta = $environ2int("DELTA", 1);
13 $display("top.v t=%d: back from environ2int with iter=%d delta=%d",
14 $time, iterations, delta);
15 end
16
17 // Stimulus generator : run 'iter' iterations
18 initial begin
19 Clock = 1'b0;
20 Reset = 1'b1;
21 iter = 0;
22 forever begin
23 #5 Clock = ~Clock;
24 #5 Clock = ~Clock;
```

Step (F11) does a "cross-step" from Vlog into C

DVE - TopLevel.1 - [Source.1 - Top: top.v]

File Edit View Simulator Signal Scope Trace Window Help

0 x 1s

1 module Top;

```
2
3 reg Clock, Reset;
4 integer iter; // number of iterations
5 integer delta; // increment between two steps
6
7 // use PLI function to get number of iterations
8 reg[7:0] iterations;
9 initial begin
10 $display("top.v t=%d: about to call environ2int", $time);
11 iterations = $environ2int("ITERATIONS", 40);
12 delta = $environ2int("DELTA", 1);
13 $display("top.v t=%d: back from environ2int with iter=%d delta=%d",
14 $time, iterations, delta);
15 end
16
17 // Stimulus generator : run 'iter' iterations
18 initial begin
19 Clock = 1'b0;
20 Reset = 1'b1;
21 iter = 0;
```

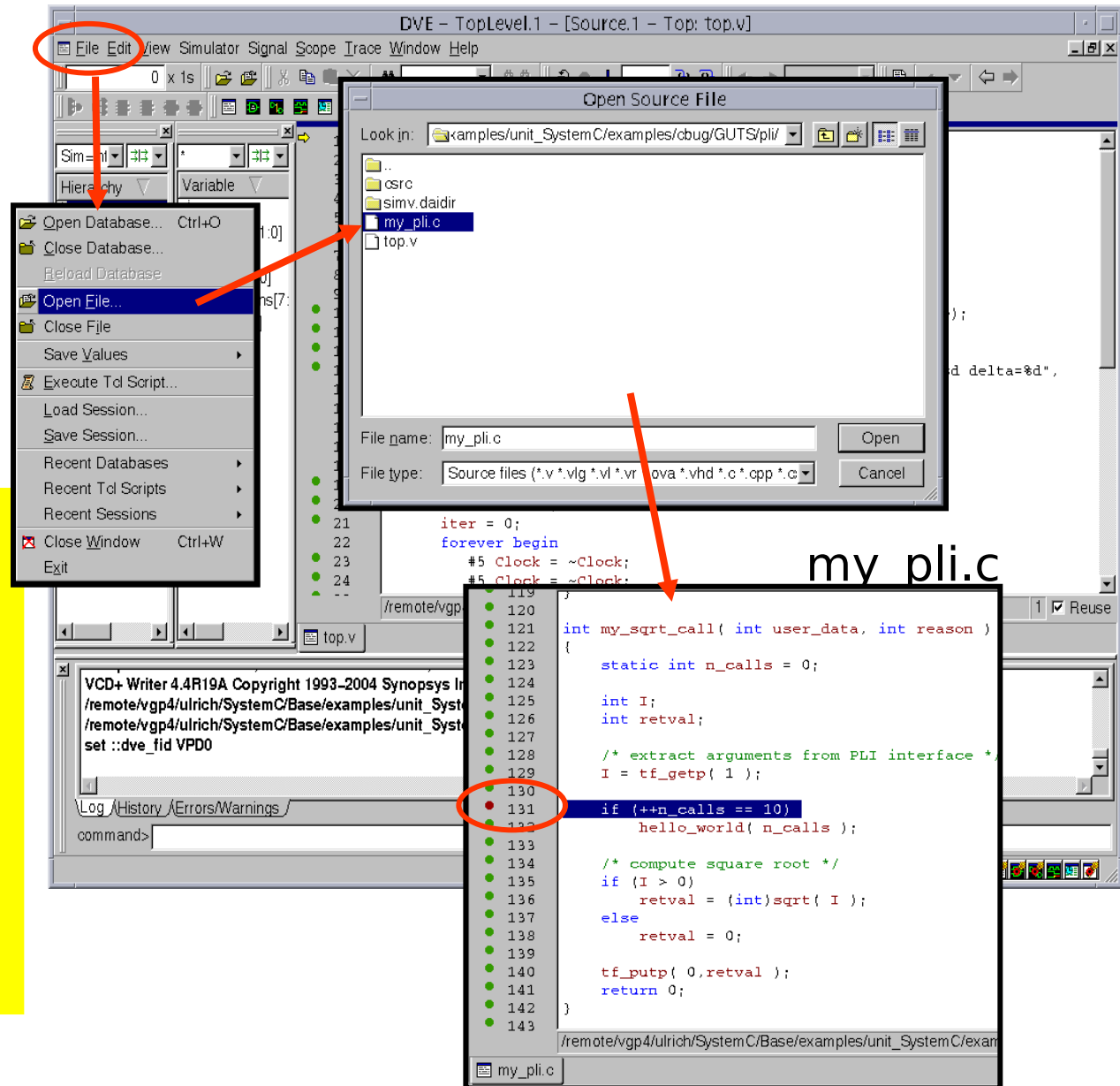
C source

```
52 if( !{acc_object_in_typelist(arg_handle, valid_types)} )
53     tf_error ("ERROR: environ2int(): second arg must be nu
54 }
55 return 0;
56
57
58 int environ2int_call( int user_data, int reason )
59 {
60     int default_value, value;
61     char *name, *s;
62
63     /* extract arguments from FFI interface */
64     name = acc_fetch_tfarg_str( 1 );
65     value = acc_fetch_tfarg_int( 2 );
66
67     /* extract value from environment */
68     value = default_value;
69     s = getenv(name);
70     if (s) {
71         value = strtol(s,0,0);
72     } else {
73         s = "-not defined-";
74     }
75     fprintf(stdout, "my_pli.c: environ2int(\"%s\",%d) => \"%s\" =>
```

There multiple ways to make DVE stop in C source code:

(B) Open C source file,
set breakpoint
and continue
simulation

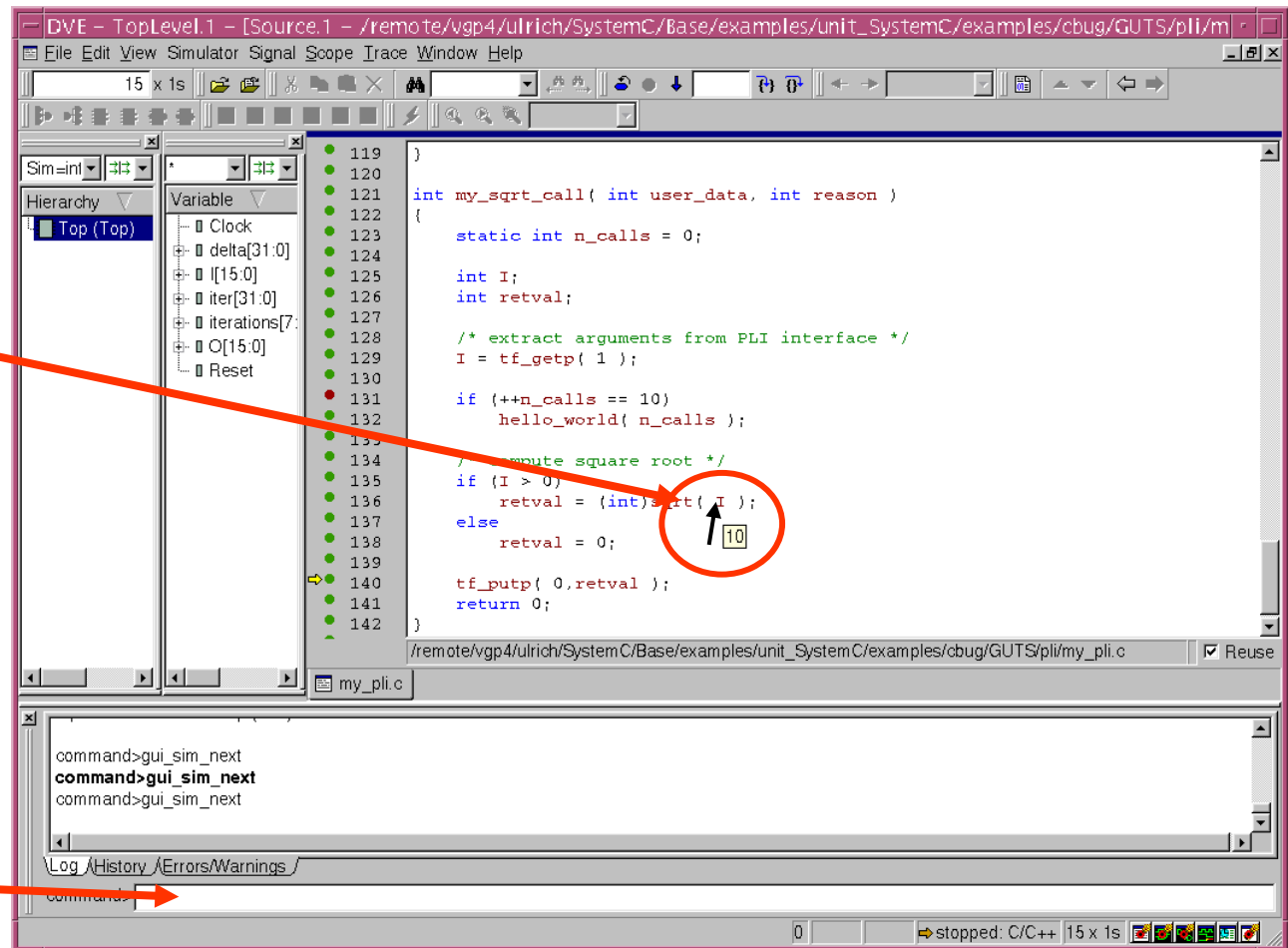
➔ Always works:
PLI, DirectC,
VhPI, MHPI, ...



How to read values from C source file?

(1) Tooltip: hold mouse over variable and wait 1 second

(2) Or enter UCLI command(s):
get retval

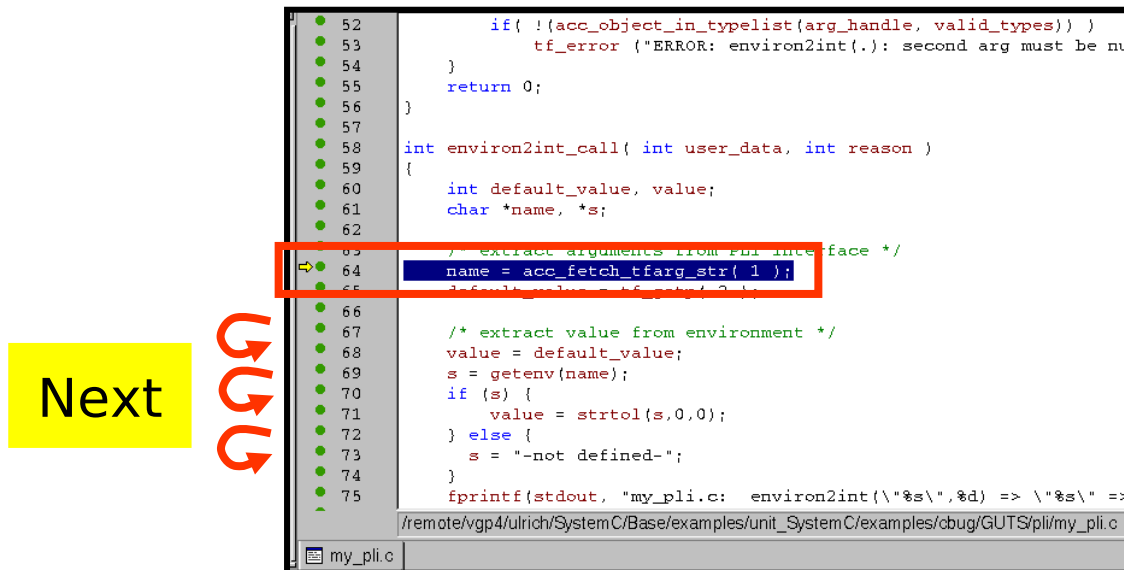


Stepping within C sources:

- **Next** goes to next statement, usually next line
- **Step** goes (if possible) into sub-functions, even system functions like printf() which can be confusing.

→ Recommendation: **always use Next in C source files**

C source



```
52         if( !{acc_object_in_typelist(arg_handle, valid_types)} )
53             tf_error( "ERROR: environ2int(.): second arg must be nu
54         }
55         return 0;
56     }
57
58     int environ2int_call( int user_data, int reason )
59     {
60         int default_value, value;
61         char *name, *s;
62
63         /* extract arguments from PLI interface */
64         name = acc_fetch_tfarg_str( 1 );
65         /* extract value from environment */
66         value = default_value;
67         s = getenv(name);
68         if (s) {
69             value = strtol(s,0,0);
70         } else {
71             s = "not defined-";
72         }
73         fprintf(stdout, "my_pli.c: environ2int(\"%s\",%d) => \"%s\" =>
74
75     }
```

At the end of PLI function, a Step or Next will bring you back to Verilog

If you accidentally stepped into system or VCS kernel:
➔ File without-g.txt will be displayed

How to get out again:

- **continue simulation** and stop at next breakpoint
- Optional: use **Simulator/Show Stack** dialog to see how you got in there
- If necessary, use **Simulator/Move Stack Up** command repeatedly to get to good source file, set breakpoint over there and then continue

